ModelIt!

Teacher's Guide

Is Your Tap Water Safe?

Modeling Water Contamination Pathways and Public Health Systems

Quick Reference

Grade Level	Grade 12L1
NGSS Standard	HS-PS1-2, HS-ESS3-4
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-PS1-2, HS-ESS3-4

Construct and revise an explanation for the outcome of a simple chemical reaction based on the outermost electron states of atoms. Evaluate or refine a technological solution that reduces impacts of human activities on natural systems.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop computational models to predict how source water quality, treatment, infrastructure age, and corrosion interact to determine tap water safety.
Disciplinary Core Idea	**PS1.A Structure and Properties of Matter / ESS3.C Human Impacts on Earth Systems** — Water chemistry (pH, dissolved minerals, chloride) determines corrosion rates in pipe materials. Human infrastructure decisions directly affect water quality at the tap.
Crosscutting Concept	**Systems and System Models** — Students analyze water safety as an emergent property of an interconnected system where failure in any component can cascade through the entire system.

CAST Testing Alignment

MULTIPLE CHOICE:

A city switches from a lake water source to a river water source with higher chloride content. The treatment plant does not adjust corrosion control chemicals. What does the model predict will happen to lead levels at the tap and why?

CONSTRUCTED RESPONSE:

Using your model, explain why replacing lead pipes is a more permanent solution to lead contamination than adjusting water chemistry. Reference the relationships between at least three components.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Every day, 300 million Americans turn on their taps and trust that the water is safe. Most of the time, it is — the United States has one of the safest public water supplies in the world. But the infrastructure that delivers that water is aging, with some pipes dating back over a century, and the consequences of system failure can be catastrophic.

How Water Treatment Works

Modern water treatment is a multi-barrier approach: coagulation removes large particles, sedimentation allows solids to settle, filtration through sand and activated carbon removes smaller contaminants, and disinfection (chlorine or UV) kills pathogens. This process removes 99.9% of harmful organisms. However, treatment at the plant doesn't prevent contamination that occurs AFTER water enters the distribution system.

The Lead Pipe Problem

Before 1986, lead was commonly used in water service lines, solder, and fixtures. The EPA estimates 6-10 million lead service lines remain in use across the US. Lead doesn't typically leach into water when pipes develop a protective mineral coating (called a scale). However, if water chemistry changes — becoming more acidic or containing more chloride — this protective coating dissolves, releasing lead directly into drinking water.

The Flint Water Crisis

In April 2014, Flint, Michigan switched its water source from Lake Huron to the Flint River to save $5 million over two years. The river water was more corrosive, but the city failed to add corrosion control chemicals (orthophosphate). Within months, the protective scale inside Flint's lead pipes dissolved, and lead levels at some taps exceeded 13,000 ppb — nearly 900 times the EPA action level. An estimated 6,000-12,000 children were exposed to elevated lead, causing irreversible neurological damage.

Environmental Justice and Water Access

Water contamination disproportionately affects communities of color and low-income communities. EPA data shows that drinking water violations are 40% more likely in communities with higher poverty rates. These communities often have the oldest infrastructure, least political influence, and fewest resources for alternatives. The Flint crisis highlighted how water safety is not just a technical issue but an issue of equity and justice.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify source water quality, treatment effectiveness, pipe infrastructure age, corrosion rate, and tap water safety as the key components of the drinking water system.

E **Establish Relationships**
Students determine that source chemistry and pipe age drive corrosion rate, treatment mitigates source contamination, and the interaction of all factors determines final tap water safety.

V **Visualize & Model**
Students build a computational model showing how the water delivery system functions as an interconnected chain where failure at any point can compromise safety at the tap.

E **Evaluate Outcomes**
Students run healthy system, Flint crisis, and infrastructure neglect scenarios to identify which failures are most dangerous and why cascading failures are hardest to prevent.

R **Revise & Extend**
Students add testing frequency, community socioeconomic status, or regulatory enforcement to explore how policy and social factors interact with the physical system.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Today we're investigating something that affects every single one of you. Modeling Water Contamination and Public Health."*

Watch For

Show cover image. Ask: What do you already know about this topic? Quick share-out.

Slide 2: Learning Objectives

3 min

Say: *"Here's what you'll be able to do by the end of today's lesson."*

Watch For

Have students read objectives aloud. Pre-teach vocabulary with quick visual aids.

Slide 3: Big Question

3 min

Say: *"This is our driving question. By the end of class, you'll be able to answer it with evidence from your model."*

Watch For

Quick-write: Students write their initial hypothesis. Save to compare later.

Slide 4: LEVER Framework

2 min

Say: *"We're going to build a computational model to investigate this system. Here are our five steps."*

Watch For

Preview each LEVER step. Emphasize that modeling reveals hidden patterns.

Slide 5: Activity 1: Components

8 min

Say: *"What are the key parts of this system? Which ones come from outside, and which respond inside?"*

Watch For

Guide sorting of external vs. internal components. Use think-pair-share for reasoning.

Slide 6: Activity 2: Connections

8 min

Say: *"In Flint, switching water sources increased corrosion. Why didn't treatment prevent the crisis?"*

Watch For

Students draw arrows showing +/- relationships. Discuss feedback loops if present.

Slide 7: Activity 3: Simulation

10 min

Say: *"Now let's test our model. Run each scenario and record what happens."*

Watch For

Students run all scenarios, record predictions vs. observations. Discuss surprises.

Slide 8: Discoveries

5 min

Say: *"So what did our model reveal? Your tap water depends on source protection, treatment, and distribution all working together...."*

Watch For

Class discussion of discoveries. Compare initial hypotheses to model evidence.

Slide 9: STEM Challenge

12 min

Say: *"Now apply what you've learned. Design a low-cost monitoring system for early contamination detection...."*

Watch For

Groups design solutions using model data. Present and evaluate designs.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Source Water Quality, Pipe Infrastructure Age
- **INTERNAL:** Treatment Effectiveness, Corrosion Rate, Tap Water Safety

Relationships

- **Source Water Quality to Treatment Effectiveness:** POSITIVE (+) — Better source water quality makes treatment more effective because the plant has fewer contaminants to remove, staying within its design capacity.
- **Source Water Quality to Corrosion Rate:** NEGATIVE (-) — Better source water quality (proper pH, low chloride) reduces pipe corrosion. Corrosive source water attacks pipe materials and dissolves protective coatings.
- **Pipe Infrastructure Age to Corrosion Rate:** POSITIVE (+) — Older pipes with thinner protective coatings or deteriorating materials corrode faster, especially when exposed to changes in water chemistry.
- **Corrosion Rate to Tap Water Safety:** NEGATIVE (-) — Higher corrosion releases more heavy metals (lead, copper) into the water between the treatment plant and the tap, directly reducing safety.

Key Discoveries

- Water safety is a SYSTEM property — no single component guarantees safe water; the system depends on source quality AND treatment AND distribution integrity working together
- Corrosion control is the critical link — when water chemistry changes without adjusting corrosion treatment, protective pipe coatings dissolve and heavy metals leach into drinking water
- Aging infrastructure is a ticking time bomb — even with excellent source water and treatment, deteriorating pipes can introduce contaminants after the water leaves the plant
- The Flint crisis was a cascading system failure — changing the source without adjusting treatment for the new water chemistry triggered pipe corrosion that poisoned an entire city

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"If water is clear, it's safe to drink"	Many dangerous contaminants are invisible, odorless, and tasteless. Lead, PFAS, arsenic, and many bacteria cannot be detected by human senses. Clear water can contain lead at 100x the safe level without any visible sign. This is why regular testing at the tap — not just at the treatment plant — is essential.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"The government tests water so it must be safe"	While utilities are required to test water, testing has significant gaps. The EPA's Lead and Copper Rule only requires testing at a sample of homes, and utilities have historically selected homes less likely to show problems. Flint's water tested 'safe' for over a year because of sampling manipulation. Testing frequency, sample site selection, and reporting transparency all matter.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Bottled water is always safer than tap water"	Bottled water is regulated by the FDA with less stringent standards than the EPA applies to tap water. Studies have found that 25% of bottled water is simply repackaged tap water, and some brands have tested positive for microplastics and contaminants. Tap water in most US cities is tested hundreds of times per month, while bottled water may be tested only weekly.